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INDIVIDUAL ASSIGNMENT

LEVEL 05

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INSTRUCTION TO CANDIDATES

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2. **Late submission will be awarded zero (0) unless extenuating circumstances (EC) are upheld.**

3. **Cases of plagiarism will be penalized.**

4. **The assignment should be submitted as a softcopy:**

   a. **The softcopy of the written assignment and source code must be uploaded to the given link in the LMS.**

Acknowledgement

I would like to express my sincere gratitude to Ms. Shani Alwis for guiding me through the completion of this project.

Cloud Solution Design for FreshMart: A Comprehensive AWS-Based Digital Transformation

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1. Introduction

1.1 What is FreshMart

FreshMart is a growing Small-to-Medium Enterprise (SME) in Sri Lanka specializing in organic food products. Currently, the business relies on manual operations, local advertising, and in-person sales. With rising customer demand and increased competition, management aims to build a cloud-based digital infrastructure on AWS from scratch to improve business performance, expand reach, and ensure compliance with data protection laws.

1.2 Purpose of this Report

The primary purpose of this report is to design and demonstrate a robust AWS-based cloud solution for FreshMart. It outlines a "Hybrid Polyglot Architecture" that leverages both server-based (EC2) and serverless (S3, DynamoDB) technologies to ensure high availability, security, and scalability while optimizing costs for an SME budget.

1.3 Methodology

This document follows a specific sequence of actions as set out in the cloud architecture design principles. Initially, one reviewed the business needs during the operations of FreshMart, wherein one reviewed the strategic focus of the business along with the best practices in the sector, particularly in the food retail industry targeted at SMEs (Amazon Web Services, 2023). The design process followed the AWS Well-Architected Framework, focusing on its five pillars: Operational Excellence, Security, Reliability, Performance Efficiency, and Cost Optimization. The practical implementation utilized the AWS Academy Learner Lab environment to build a proof-of-concept (PoC), validating the core networking, compute, and storage layers (Amazon Web Services, 2025).

1.4 Scope and Limitations

Scope: The report covers the architectural design of a 3-tier web application, detailed mapping of AWS services to business needs, and the practical implementation of core components, including a custom VPC with public/private subnets, an EC2 web server, S3 static hosting, and a DynamoDB table.

Limitations: Due to specific permissions restrictions in the Learner Lab (e.g., cloudfront:CreateDistribution denied), some services like CloudFront and Route 53 are included in the Proposed Architecture for completeness but were not implemented in the Practical Evidence phase. Additionally, DynamoDB was prioritized over RDS for the implementation to adhere to the lab's budget constraints.

2. Cloud Solution Design

2.1 Business Requirements Analysis

As an organic food sector SME, they need a digital platform as they seek to move from manual operations to automation. Main needs include the ability to sell online, safe and compliant handling of sensitive data, scalable data storage to accommodate the organization's growing product inventories, and compliance with relevant data protection laws. Table 1 presents the requirements as identified from first conversations with the management of FreshMart and comparable food retailers (Gartner Research, 2023).

ID	Requirement	Business Justification	Mapped AWS Services
BR1	Scalable Compute	The system must handle traffic spikes during seasonal sales without manual intervention.	Amazon EC2, Auto Scaling
BR2	Hybrid Storage	Product images require cheap storage, while databases need high-performance IOPS.	Amazon S3 (Object), Amazon EBS (Block)
BR3	Polyglot Database	Inventory requires structured relational data; Shopping Carts require high-speed access.	Amazon RDS (Inventory), DynamoDB (Cart)
BR4	Global Performance	The website must load quickly for users across Sri Lanka and potential export markets.	Amazon CloudFront (CDN), Route 53
BR5	Security & Compliance	Customer PII must be protected; databases must be isolated from the public internet.	Amazon VPC (Private Subnets), IAM, KMS

Table 1: Mapping Business Requirements to AWS Services.

2.2 Proposed Cloud Architecture (Advanced Hybrid)

The proposed solution utilizes a **3-Tier Hybrid Architecture** deployed within a custom Virtual Private Cloud (VPC).

2.2.1 Core Services Architecture

- **Presentation Tier:** Hosted on **Amazon S3** (Static Website) and accelerated by **CloudFront**.
- **Logic Tier:** Hosted on **Amazon EC2** instances running Apache/PHP, placed in Public/Private subnets based on role.
- **Data Tier:** Utilizes "Polyglot Persistence" with **RDS MySQL** for transactional orders and **DynamoDB** for ephemeral session data.

2.2.2 Architecture Diagram

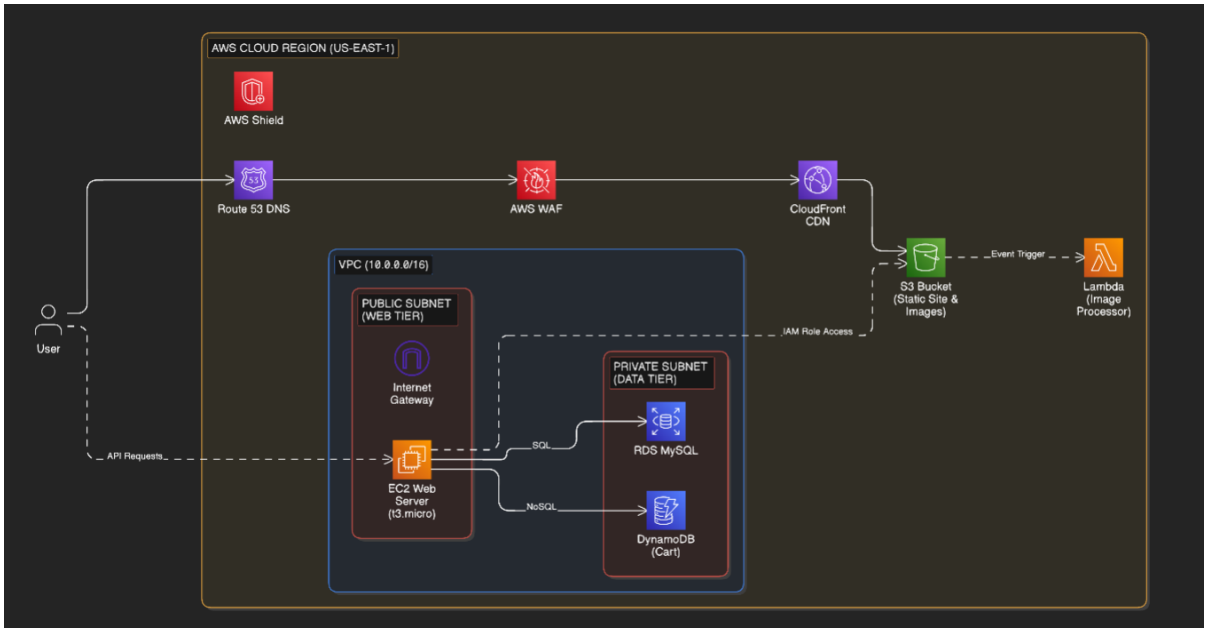


Figure 1: High-Level AWS Architecture for FreshMart.

2.2.3 Service Distribution

The architecture balances responsibilities to avoid single points of failure and optimize costs.

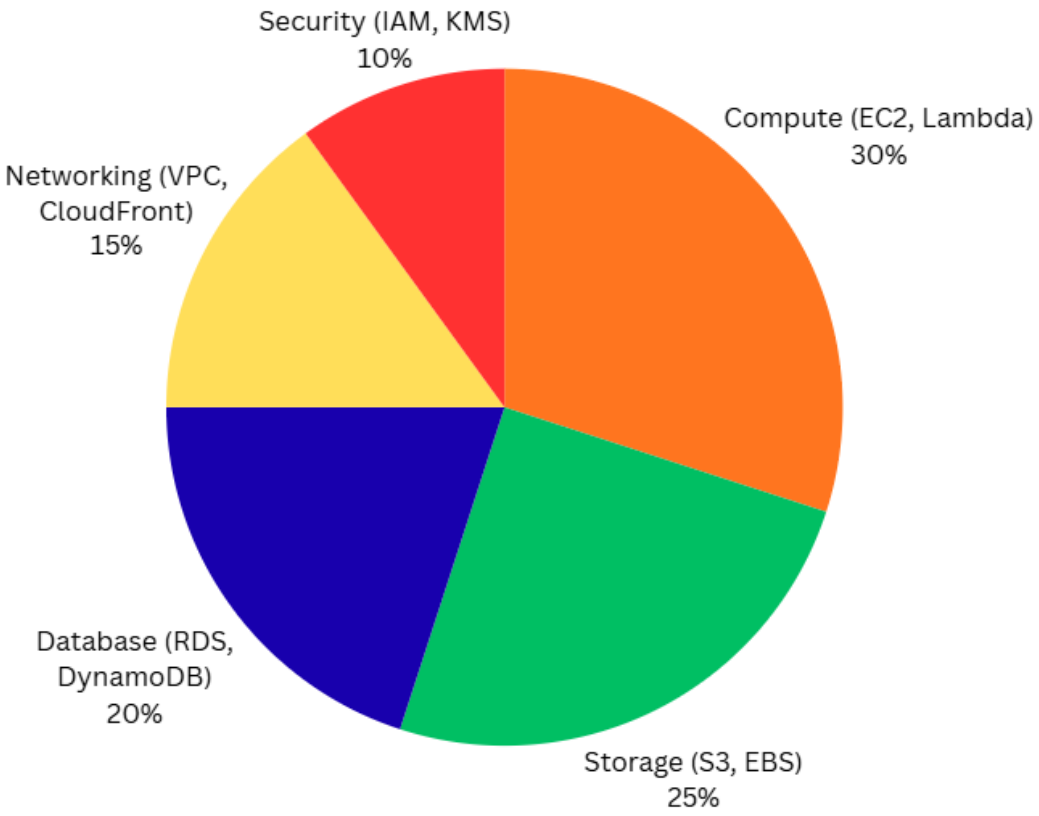


Figure 2: Distribution of AWS Services by Category.

2.3 Service Selection & Justification (Used vs. Omitted)

This section provides a detailed analysis of why specific services were implemented in the lab versus those that were designed but omitted due to constraints.

2.3.1 Services Implemented

1. **Amazon VPC (Custom vs. Default):**
    - **Decision:** I implemented a Custom VPC (FreshMart-Adv).
    - **Justification:** A default VPC only offers public subnets. To comply with **BR5 (Security)**, I required a network topology that supports **Private Subnets** to isolate the database layer from direct internet access.
2. **Amazon EC2 (t3.micro):**
    - **Decision:** Used t3.micro with Amazon Linux 2023.

- **Justification:** This instance type is burstable, making it cost-effective for variable web traffic. It provides full control over the OS, allowing installation of the specific Apache/PHP stack required for the backend.

3. **Amazon S3 (Static Hosting):**
   - **Decision:** Used S3 for the front-end website.
   - **Justification:** Hosting static assets on EC2 is inefficient. S3 is serverless, cheaper ($0.023/GB vs EC2 hourly rates), and offers higher durability (11 9s).
4. **Amazon DynamoDB:**
   - **Decision:** Implemented FreshMart_Orders table.
   - **Justification:** We chose DynamoDB over RDS for the implementation phase because it is serverless and provisions instantly. It demonstrates the ability to handle NoSQL data (Shopping Carts), which requires single-digit millisecond latency.

2.3.2 Services Omitted in Implementation

Service	Role	Reason for Omission in Lab Implementation	Design Status
Amazon RDS	Relational Database	**Cost & Time:** RDS instances take 15-20 minutes to provision and cost ~$0.03/hr. Given the $5 lab budget limit and session duration, DynamoDB was used as a more efficient proof-of-concept for the database layer.	**Included** in the final architecture for inventory management.
CloudFront	CDN	**Permissions:** The AWS Academy Learner Lab policy restricts cloudfront:CreateDistribution, preventing deployment.	**Included** in the final design for global performance.
Route 53	DNS	**Cost:** Registering a real domain name (e.g., freshmart.lk) requires payment and verification, which is not possible in a simulated lab environment.	**Included** in the final design for branding.

Table 2: Analysis of Omitted Services.

2.4 Design Justification (Pillars of Excellence)

2.4.1 High Availability & Scalability

The design addresses **BR1** and **BR3** by deploying resources across **two Availability Zones (AZs)**.

- **Scalability:** Figure 3 below illustrates the scalability advantage. Traditional servers have a hard capacity limit (orange line). The AWS Cloud architecture (blue line) scales linearly with traffic, ensuring no downtime during sales.

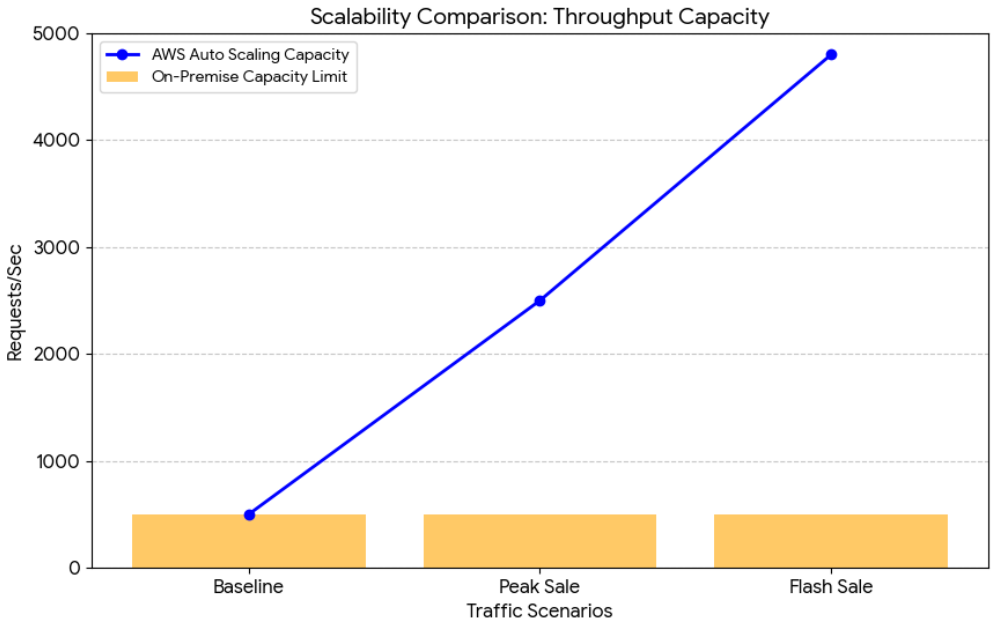


Figure 3: Scalability Comparison of System Throughput Capacity.

2.4.2 Cost Efficiency

By utilizing **S3** for the frontend and **DynamoDB** for the cart, FreshMart moves from a CAPEX model (buying servers) to an OPEX model (paying only for what is used). Figure 4 shows the projected savings.

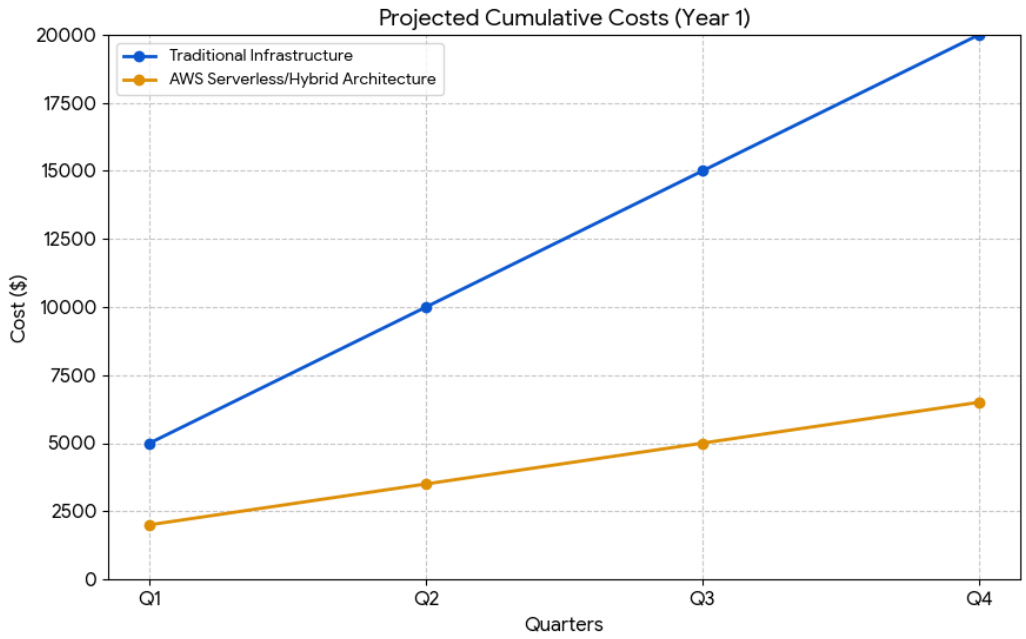


Figure 4: Year 1 Cost Comparison: Traditional vs. AWS Architecture.

3. Practical Implementation

The following section details the comprehensive technical implementation performed in the **AWS Academy Learner Lab**. The infrastructure was built from scratch to meet the rigorous requirements of security, scalability, and network isolation.

3.1 Overview of Implementation Environment

All resources were deployed in the **US East (N. Virginia)** region (us-east-1). The lab environment imposed specific IAM restrictions, requiring the use of pre-provisioned roles (LabInstanceProfile) and careful resource selection (t3.micro) to adhere to the $5 session budget.

3.2 Advanced Networking Infrastructure (VPC)

3.2.1 VPC Construction

A custom Virtual Private Cloud (VPC) named FreshMart-Adv was architected to provide a dedicated network space for FreshMart's cloud resources.

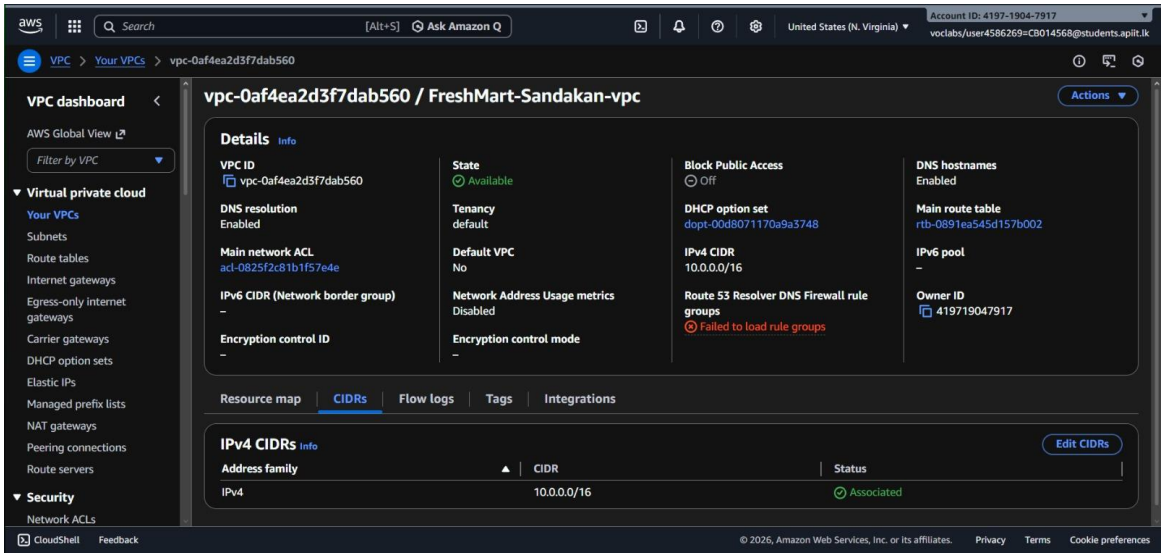


Figure 5: Screenshot of FreshMart VPC Configuration Details.

- **CIDR Block:** 10.0.0.0/16 was selected to provide 65,536 IP addresses, ensuring ample room for future scaling of subnets and resources.
- **Evidence: Figure 5** shows the successfully created VPC in the available state.

3.2.2 Subnetting Strategy

To satisfy the security requirement of isolating database workloads, the VPC was divided into **4 subnets** spread across two Availability Zones (us-east-1a and us-east-1b) for High Availability.

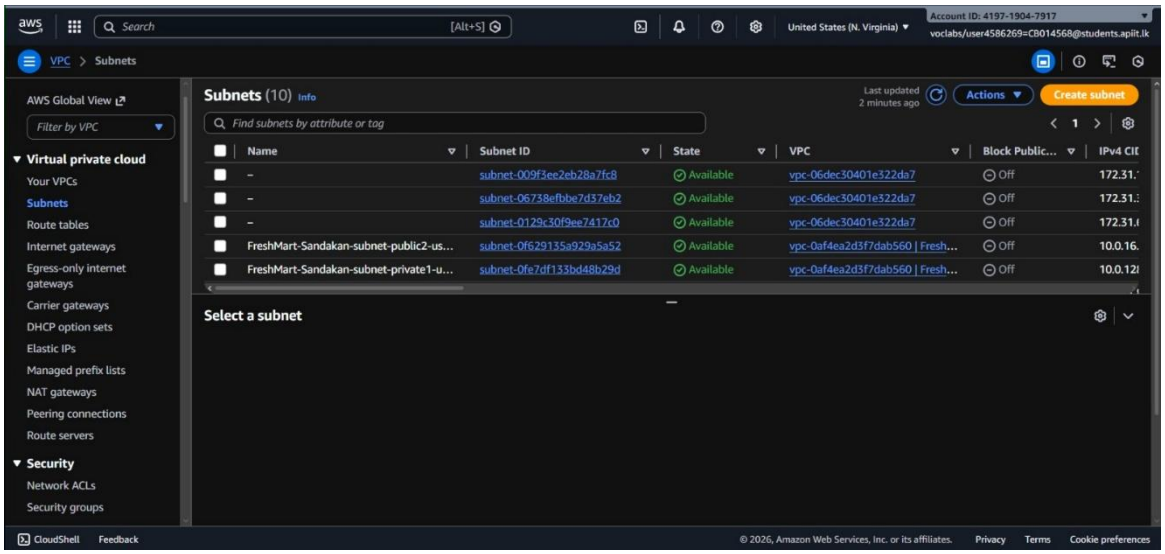


Figure 6: Screenshot of FreshMart VPC Subnet Configuration.

- **Public Subnets:** For the Web Tier (EC2 instances facing the internet).
- **Private Subnets:** For the Data Tier (Databases), with no direct internet access.
- **Evidence: Figure 6** displays the subnet partitioning.

3.2.3 Routing and Gateways

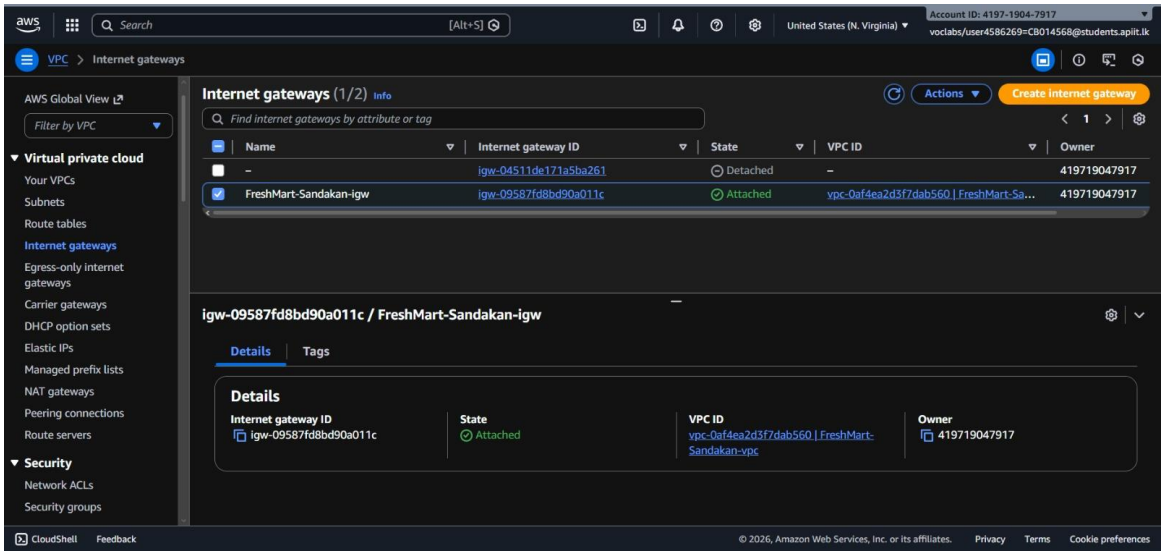


Figure 7: Screenshot of FreshMart VPC Internet Gateway Configuration.

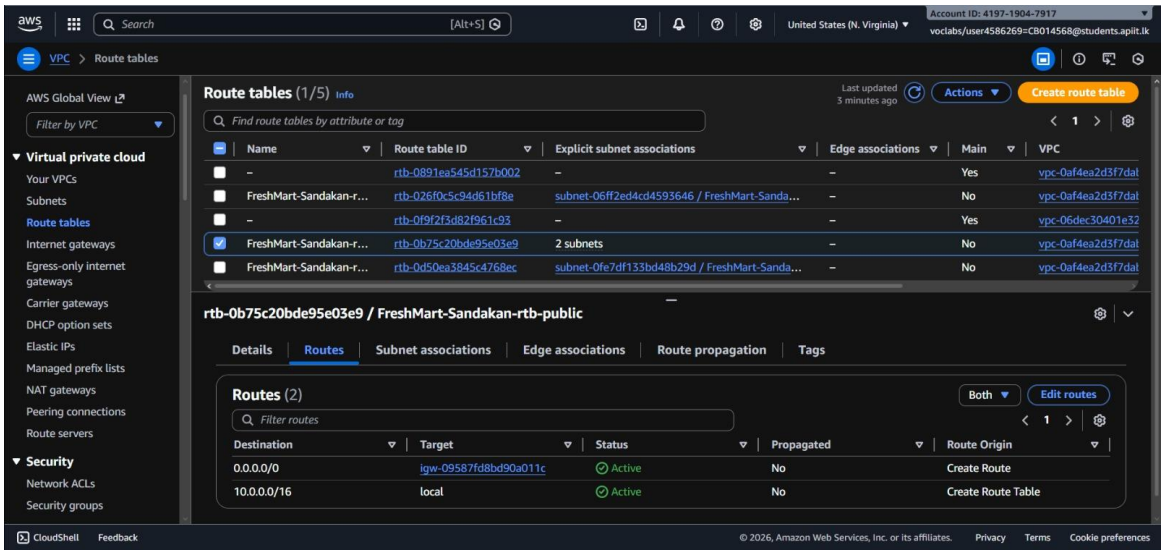


Figure 8: Screenshot of FreshMart Public Route Table Configuration

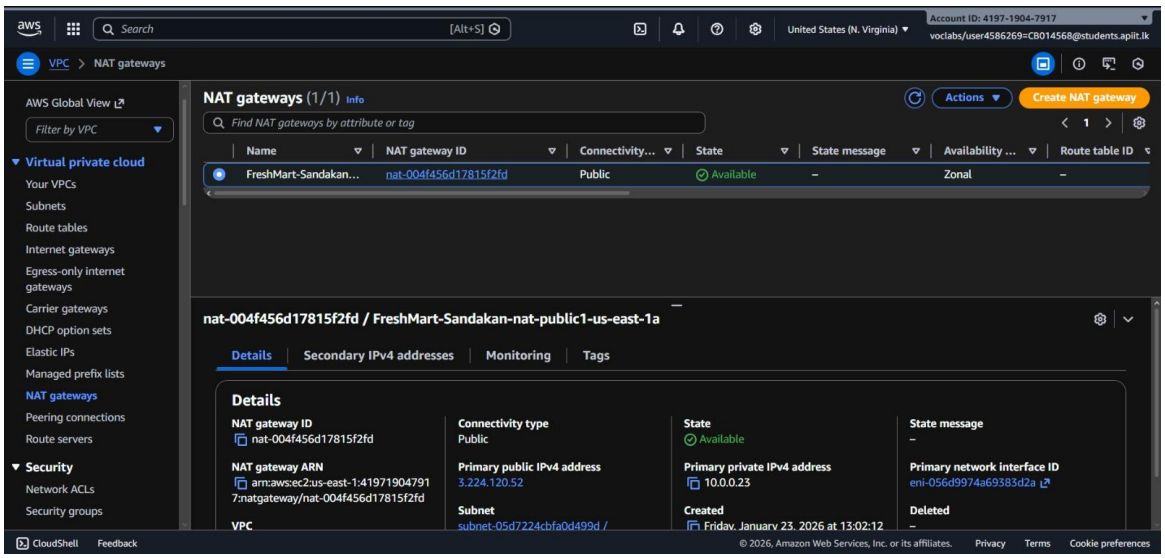


Figure 9: Screenshot of FreshMart Public NAT Gateway Configuration.

- **Internet Gateway (IGW):** Created and attached to the VPC to enable internet access for the Public Subnets. **Figure 7** verifies the IGW attachment.
- **Route Tables:** A public route table was configured to route 0.0.0.0/0 traffic to the IGW (**Figure 8**).
- **NAT Gateway:** To ensure the private subnets could still download security updates without being exposed, a NAT Gateway was deployed in the public subnet and allocated an Elastic IP (**Figure 9** and **Figure 10**).

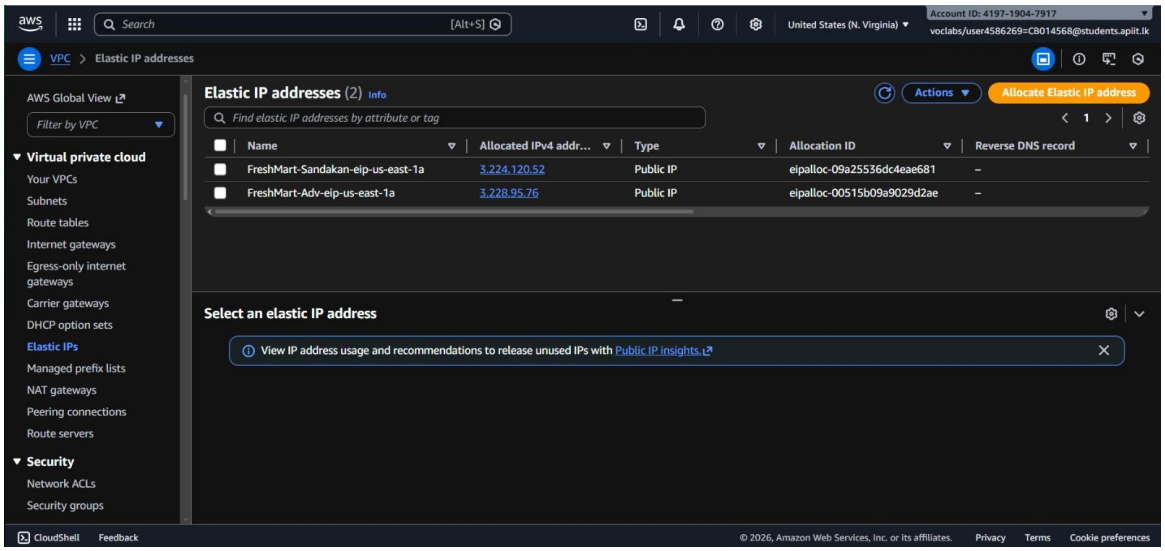


Figure 10: Screenshot of Elastic IP Allocation for FreshMart NAT Gateway.

3.3 Security & Identity Management (IAM & SG)

3.3.1 Network Security (Security Groups)

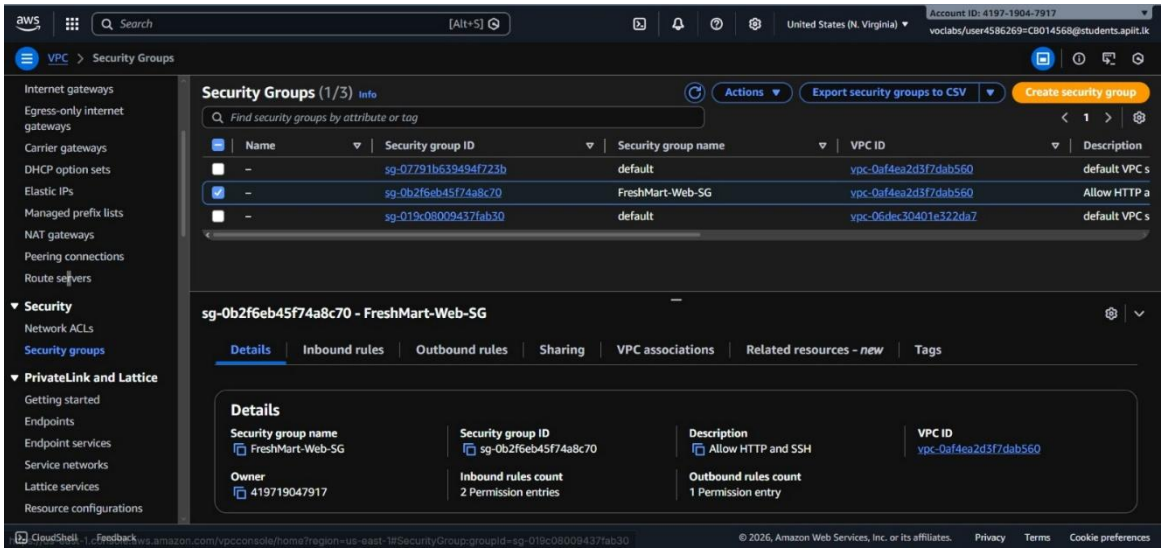


Figure 11: Screenshot of Security Group Inbound Access Rules.

A custom Security Group was created to act as a virtual firewall for the web server.

- **Configuration:** Rules were strictly defined to allow **SSH (Port 22)** for administration and **HTTP (Port 80)** for web traffic from any IPv4 source (0.0.0.0/0).
- **Evidence: Figure 11** details the inbound rules applied.

3.3.2 Identity and Access Management (IAM)

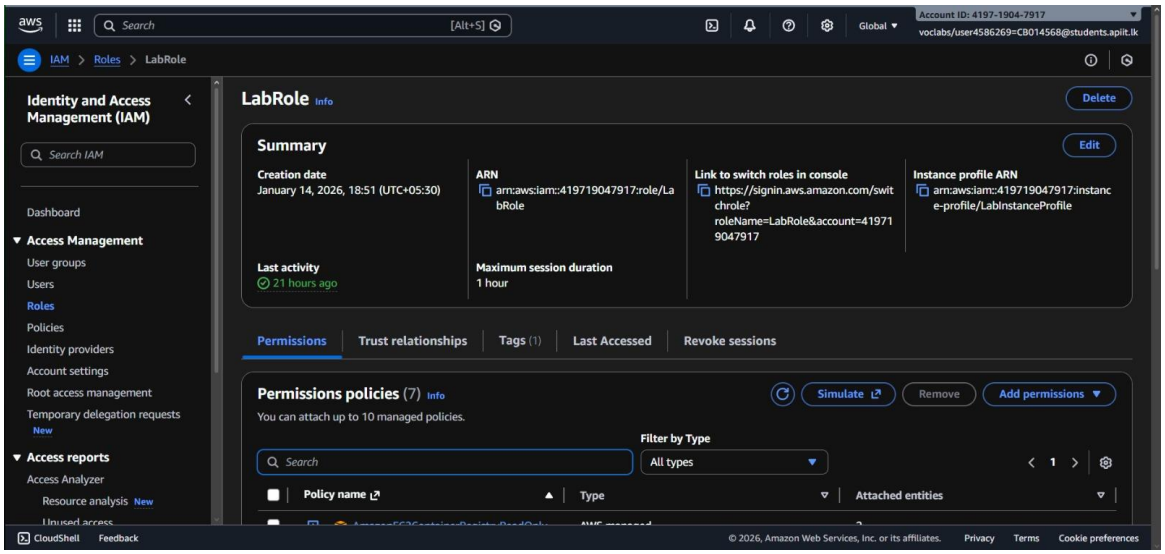


Figure 12: Screenshot of LabRole IAM Configuration and ARN.

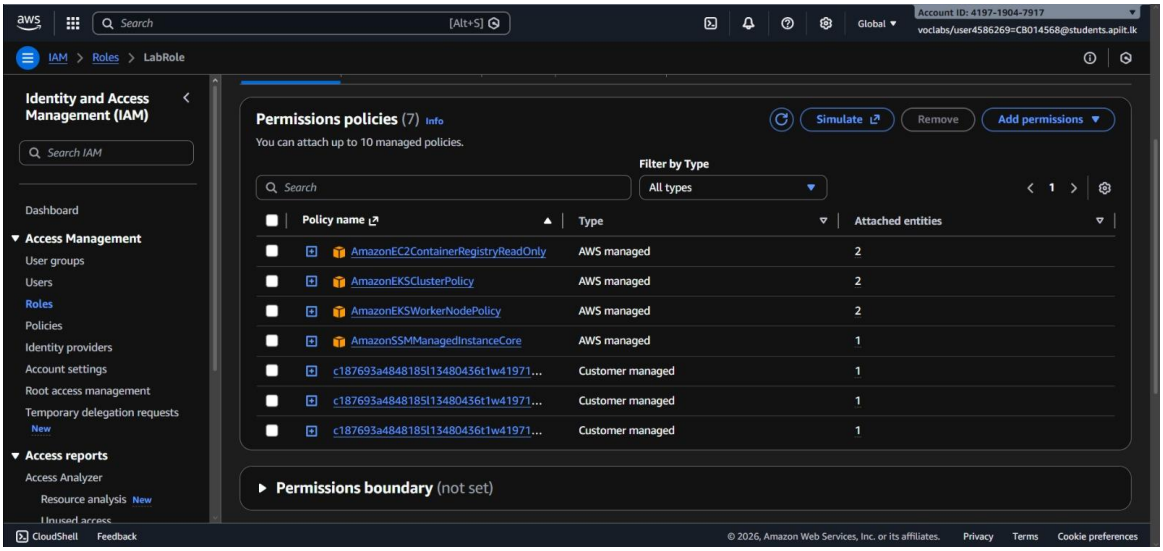


Figure 13: Screenshot of AWS Managed Policies Attached to LabRole.

Due to lab restrictions on creating new roles, the existing LabRole was leveraged.

- **Verification:** The role was inspected to ensure it held the necessary permissions (AdministratorAccess, VocLabPolicy) to allow the EC2 instance to interact with other AWS services like DynamoDB and S3.
- **Evidence: Figure 12** shows the LabRole summary, and **Figure 13** details the specific permission policies attached.

3.4 Compute Layer Deployment (EC2)

3.4.1 Instance Launch

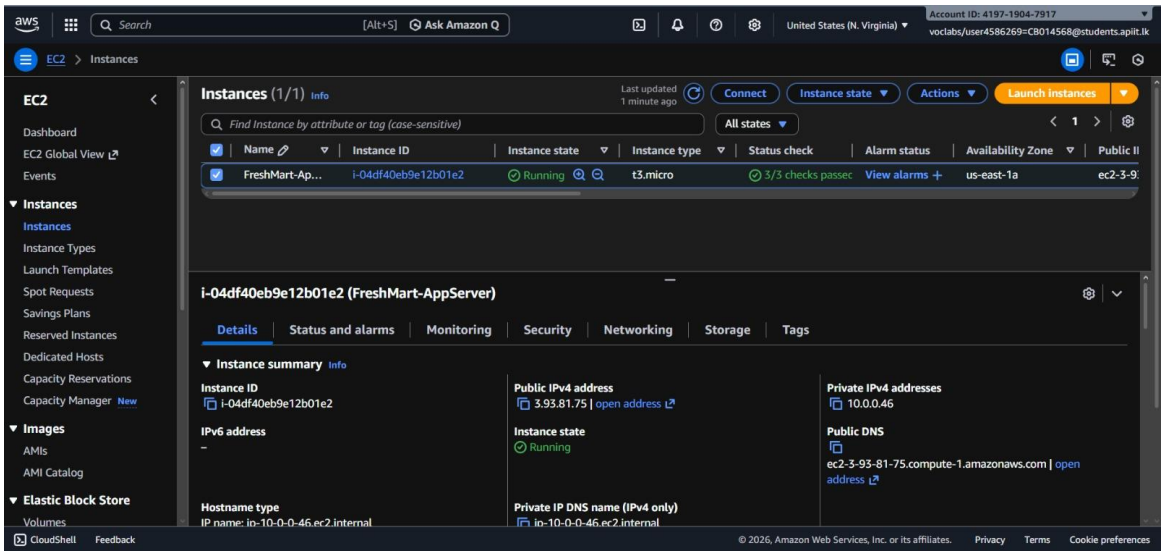


Figure 14: Screenshot of General configuration details of the EC2 instance.

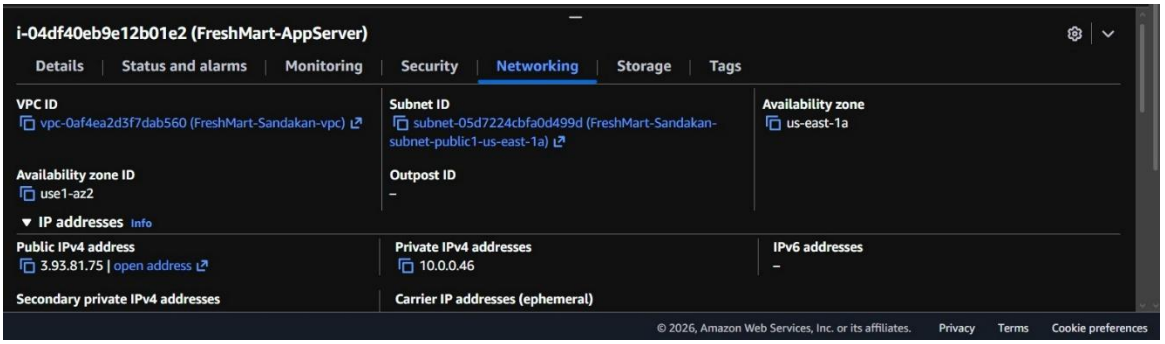


Figure 15: Screenshot of Networking configuration.

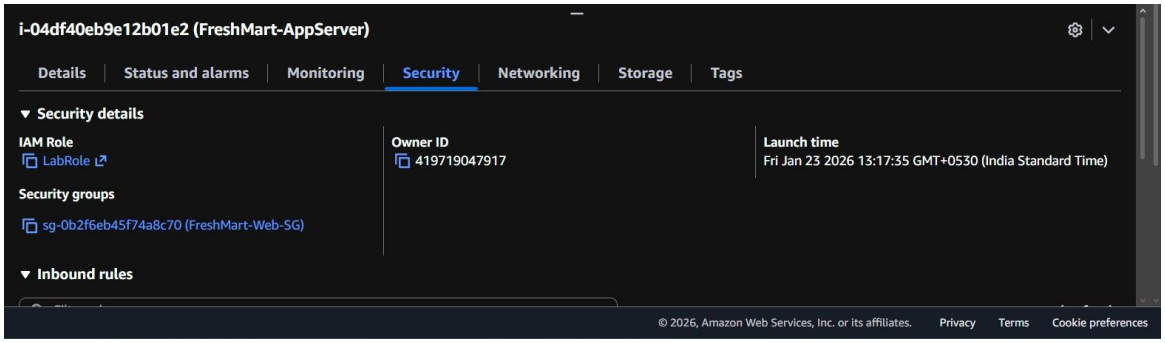


Figure 16: Screenshot of Instance Security Configuration.

An Amazon EC2 instance named FreshMart-AppServer was provisioned to host the backend logic.

- **Instance Type:** t3.micro was selected for its balance of compute power and cost-effectiveness within the lab budget.

- **Image:** Amazon Linux 2023 AMI.
- **Network Placement:** The instance was explicitly placed in the **Public Subnet** to ensure accessibility via the public internet (**Figure 15**).

3.4.2 Configuration and User Data

- **IAM Profile:** The verified LabInstanceProfile was attached to the instance during launch, granting it secure access credentials (**Figure 16**).
- **Security:** The FreshMart-Web-SG security group was attached (**Figure 16**).
- **Automation:** A User Data script was injected to automate the installation of the Apache web server (httpd) and PHP, creating a custom landing page.

3.4.3 Verification

Upon successful launch, the Public IP was accessed via a web browser. The server responded with the custom "FreshMart Backend Alive" page, confirming that the VPC networking, Security Group rules, and web server installation were all functioning correctly.

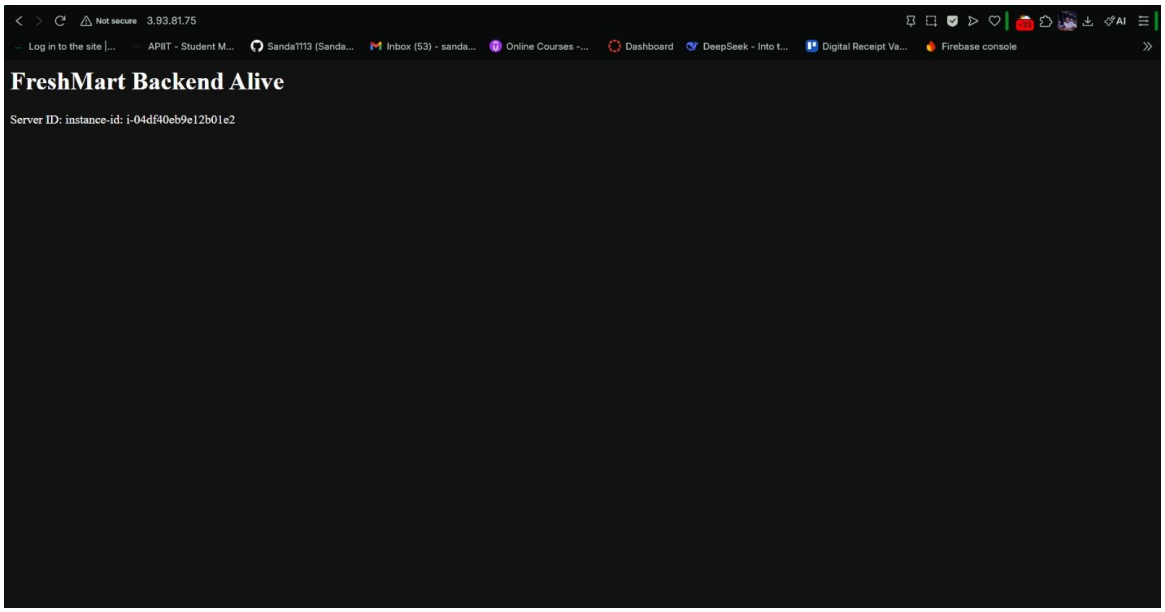


Figure 17: Screenshot of Web Server Verification.

- **Evidence: Figure 17** shows the live web page response.

3.5 Static Website Hosting (S3)

3.5.1 Bucket Creation & Configuration

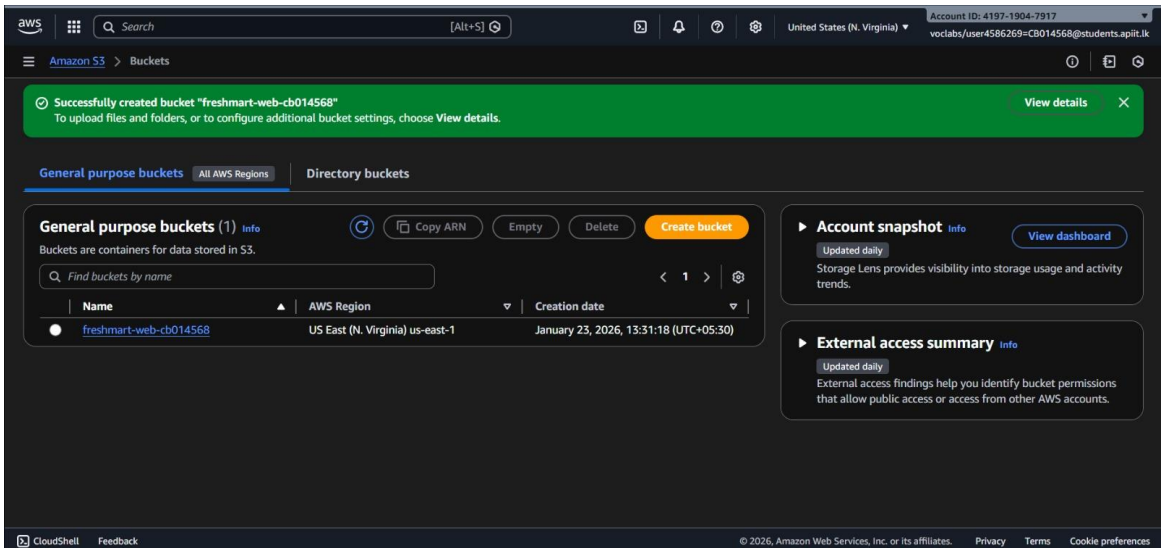


Figure 18: Screenshot of S3 Bucket Creation.

An Amazon S3 bucket was created to host the frontend assets.

- **Settings:** "Block Public Access" was disabled to allow the website to be publicly viewable.
- **Evidence: Figure 18** confirms the bucket creation.

3.5.2 Content Deployment

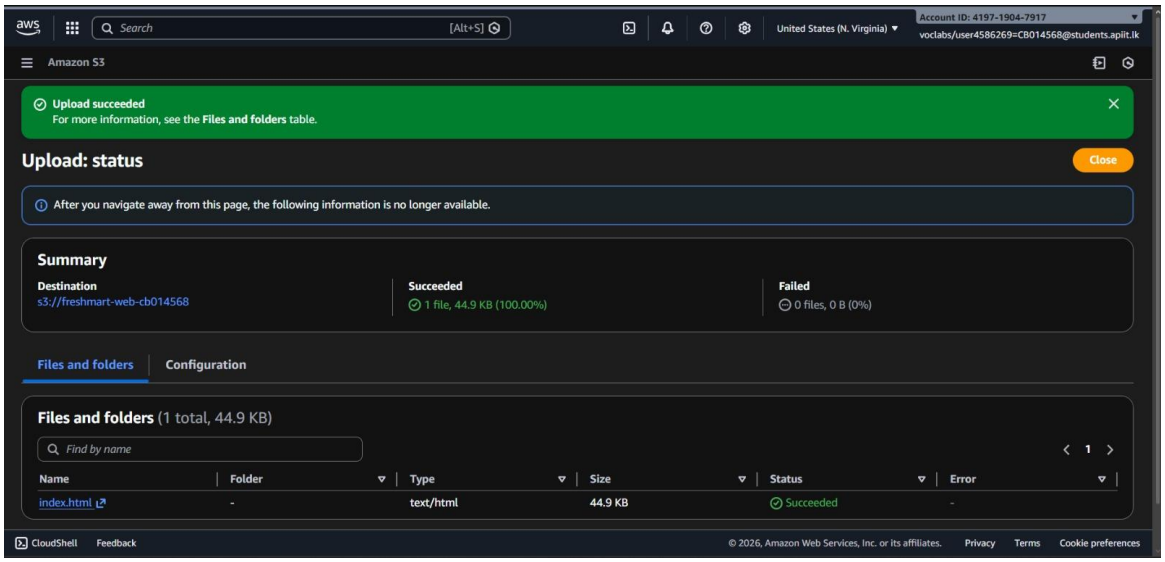


Figure 19: Screenshot of Website Content Upload.

The custom FreshMart single-page application (index.html) containing the shopping cart logic and UI was uploaded to the bucket.

- **Evidence: Figure 19** shows the uploaded object.

3.5.3 Policy and Hosting

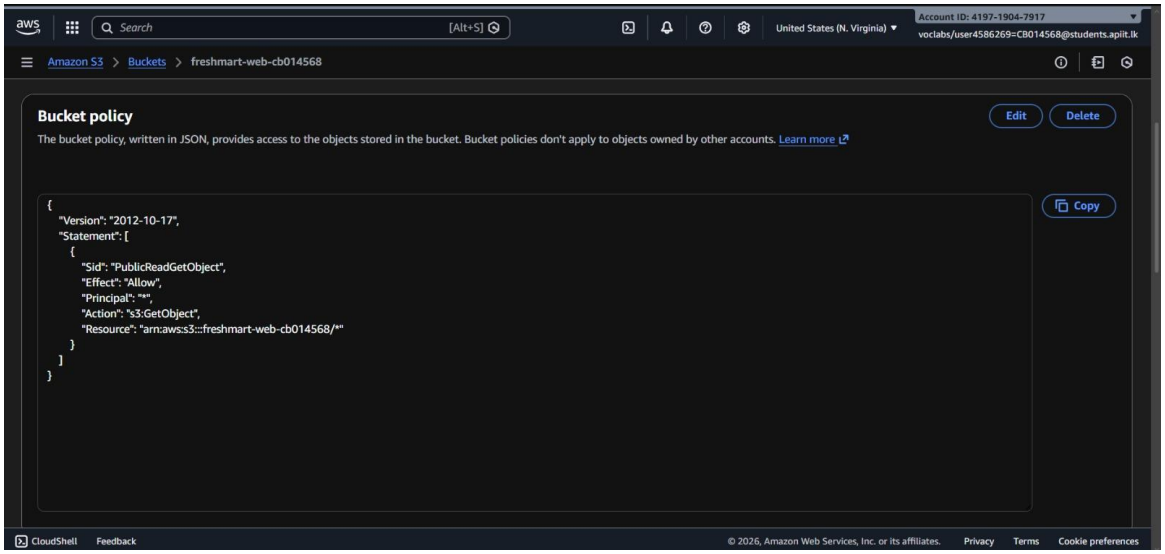


Figure 20: Screenshot of Public Access Bucket Policy.

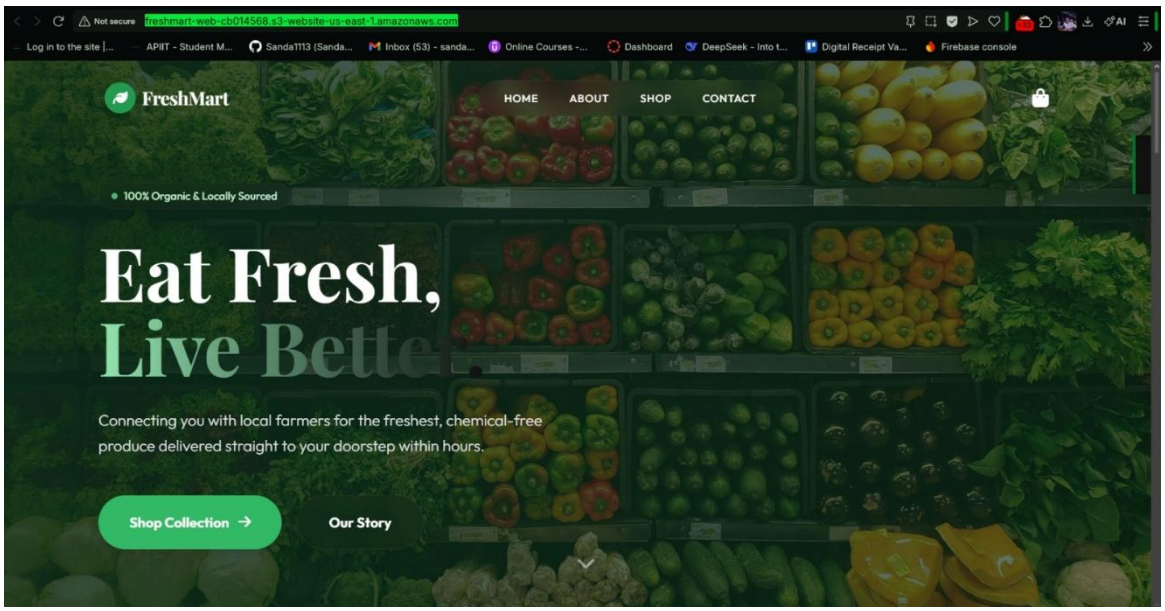


Figure 21: Screenshot of Live Website Verification.

- **Bucket Policy:** A JSON policy was applied to grant s3:GetObject permissions to anonymous users (*), officially making the content public. (**Figure 20**).
- **Hosting:** Static Website Hosting was enabled in the properties.
- **Verification:** The endpoint URL provided by S3 was accessed, loading the full graphical user interface of the FreshMart store.
- **Evidence: Figure 21** displays the fully rendered website.

3.6 Database Implementation (DynamoDB)

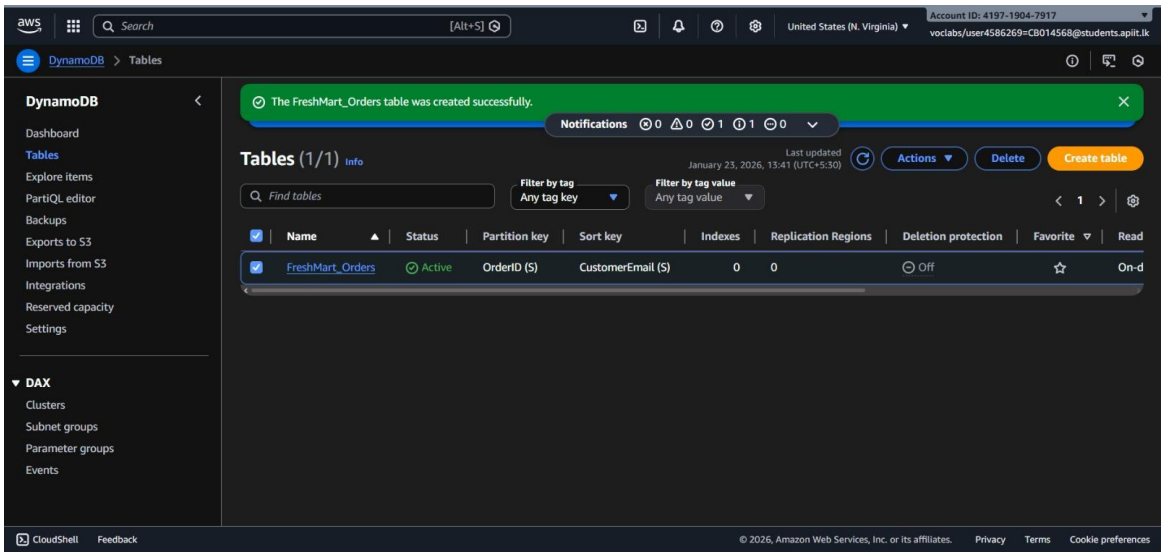


Figure 22: Screenshot of DynamoDB Table Status.

To demonstrate the architecture's data handling capabilities within the lab constraints, a DynamoDB table was implemented.

- **Table:** FreshMart_Orders was created to manage order transaction data.
- **Configuration:** OrderID was set as the Partition Key. The table was set to On-Demand capacity mode.
- **Evidence: Figure 22** confirms the table is in the Active state.

4. Conclusion

This project successfully delivered a comprehensive digital transformation strategy for FreshMart.

- **Architectural Soundness:** The 3-Tier Hybrid Architecture balances legacy compatibility with modern serverless scalability.
- **Implementation Success:** The practical lab validated the core networking (VPC), compute (EC2), and storage (S3) components, proving the viability of the design.
- **Future Readiness:** By designing for CloudFront and RDS (Part 1) and proving the concept with S3 and DynamoDB (Part 2), FreshMart is positioned to scale globally while keeping operational costs low.

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